

1. WHY GASIFICATION & ML?

Gasification turns solid waste into syngas (H_2 , CO , CO_2 , CH_4) above $700^\circ C$. H_2 is the most valuable product — it burns into clean water only.

Research Question:

Can Machine Learning accurately predict H_2 yield (vol%) from TWTS and Leather scraps gasification, using only operating conditions as inputs?

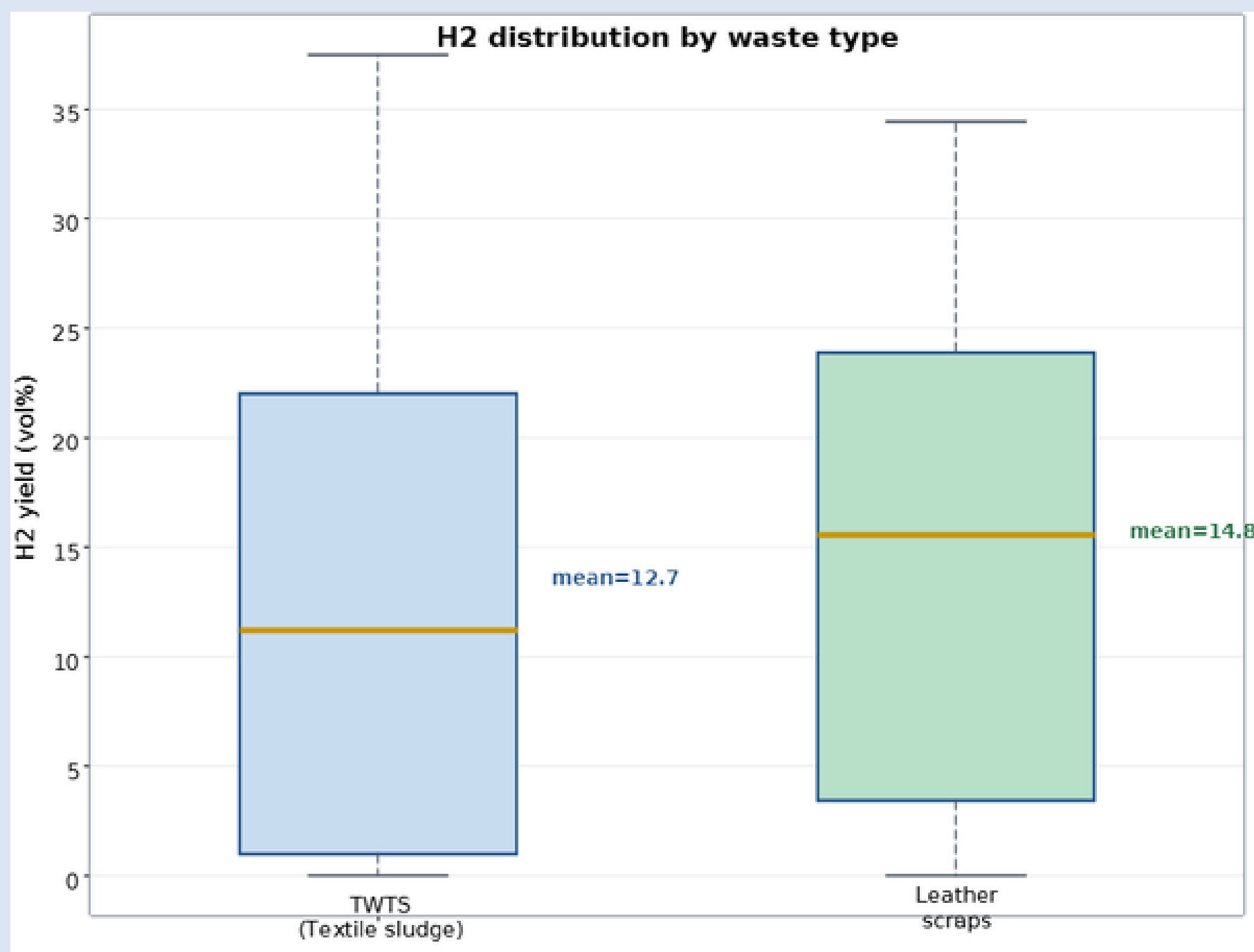
Objectives

- Compare 3 ML algorithms
- Engineer 13 thermal-history features
- Build a hybrid model per waste type
- Deploy a Streamlit prediction app

2. EXPERIMENTAL DATASET

3,748 datapoints	59 experiments
36 TWTS runs	23 Leather runs

Each run: temperature rises $\sim 15^\circ C \rightarrow 700\text{--}1000^\circ C$ target. H_2 peaks near the thermal plateau (0–37.5 vol%).



3. FEATURE ENGINEERING

9 raw variables $\rightarrow$ 24 features (+13 engineered) encoding the full thermal history.

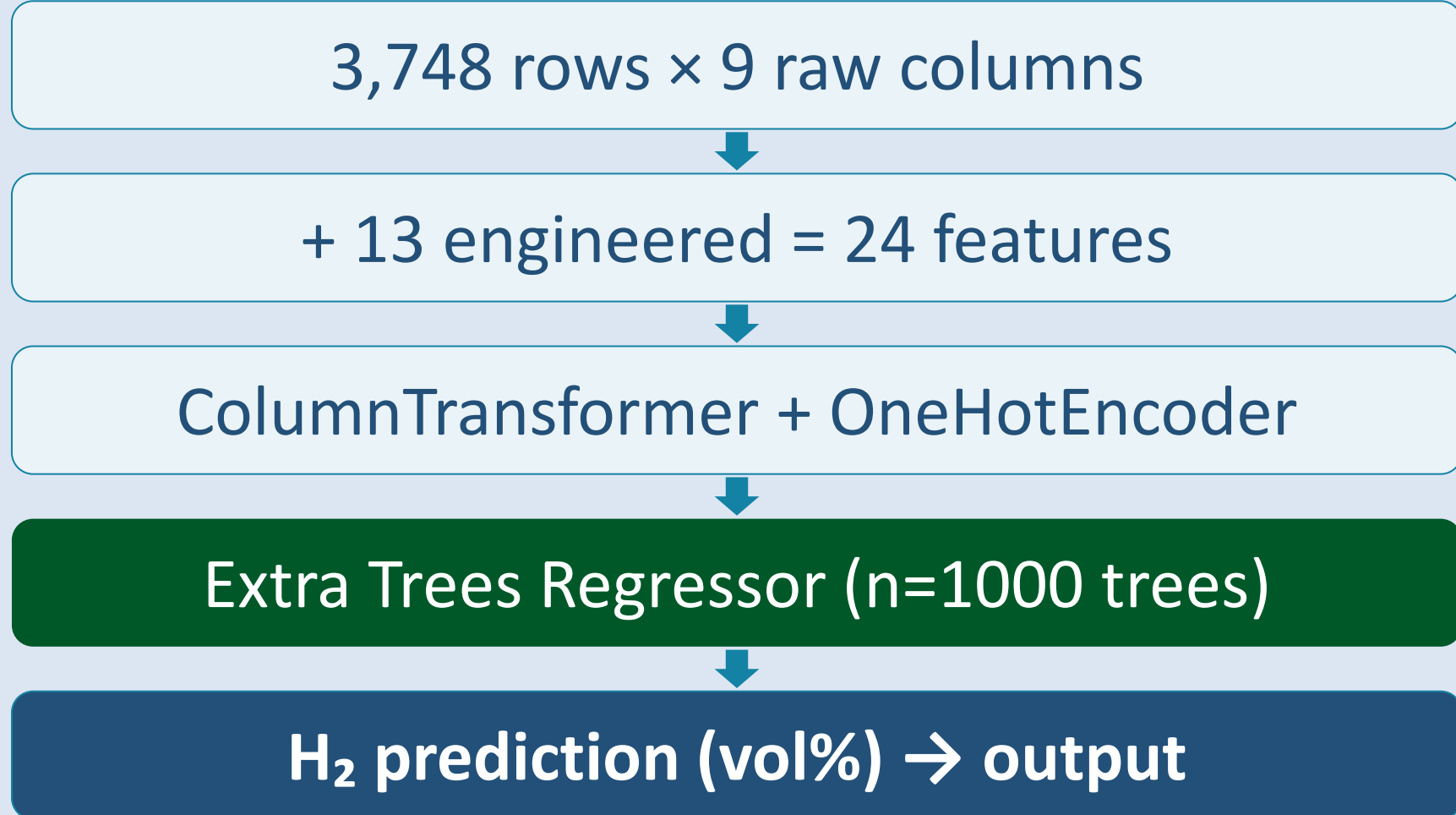
#01	Cumul_Temp	Cumulative heat energy [KEY]
#02	Thermal_Work	Temp $\times$ Time
#03	Temp_squared	Temperature nonlinearity
#04	Cumul_Temp ²	Squared cumulative energy
#05	Relative_Time	Time / max(Time) per run
#06	Thermal_Accel.	Thermal_Work / (Time+0.1)
#07	Temp_Norm	Normalized T (0-1)
#08	Temp_mean_run	Mean temp per experiment
#09	Temp_max_run	Max temp in the run
#10	Temp_std_run	Thermal variability
#11	Agent_Intensity	AgentFlow $\times$ ProcessTemp
#12	Rel_Severity	Relative_Time $\times$ ProcessTemp
#13	Kinetic_Density	AgentFlow $\times$ Relative_Time

Limitations

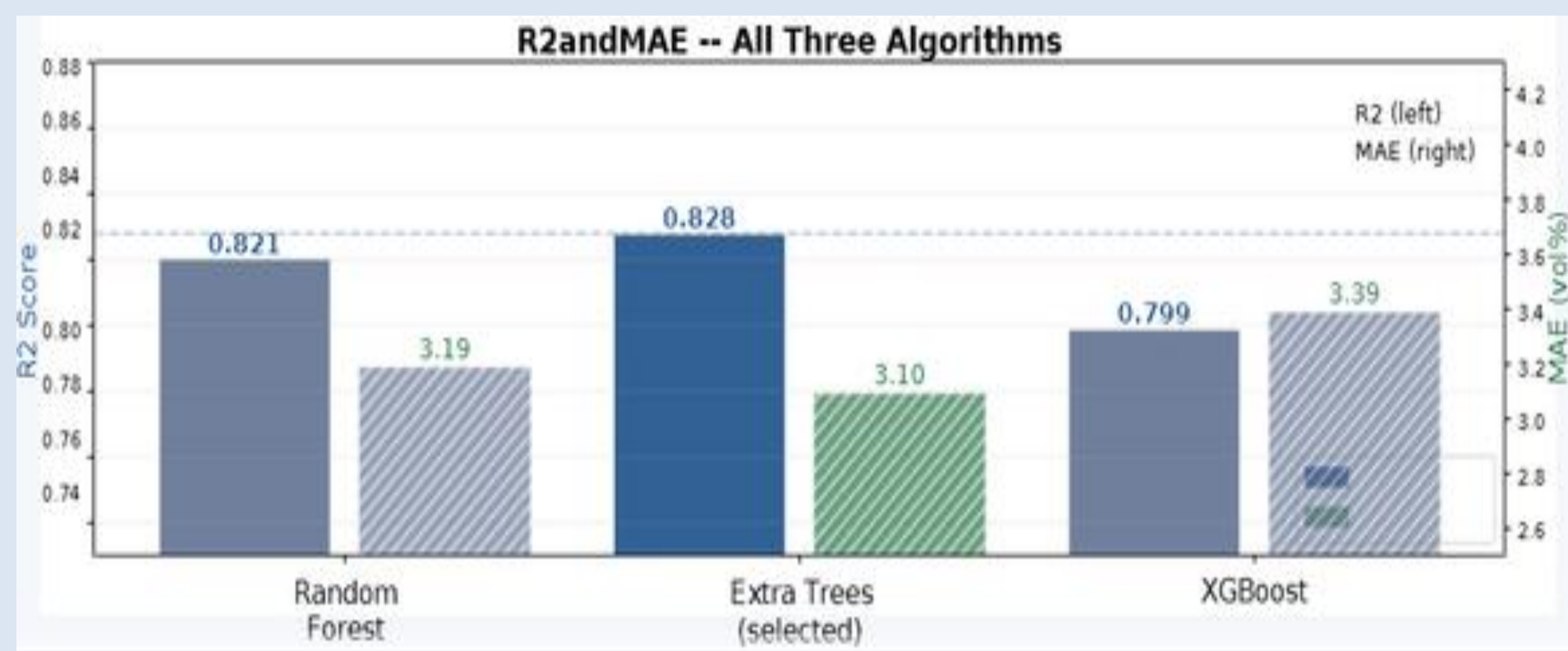
- Overfitting: R^2 train=0.974 vs test=0.828 (59 exp. only)
- Limited to fixed-bed reactor, lab-scale

4. METHODS: ML PIPELINE

Single sklearn Pipeline: features $\rightarrow$ ColumnTransformer $\rightarrow$ OneHotEncoding $\rightarrow$ regressor. Tuned with RandomizedSearchCV (30 iter).



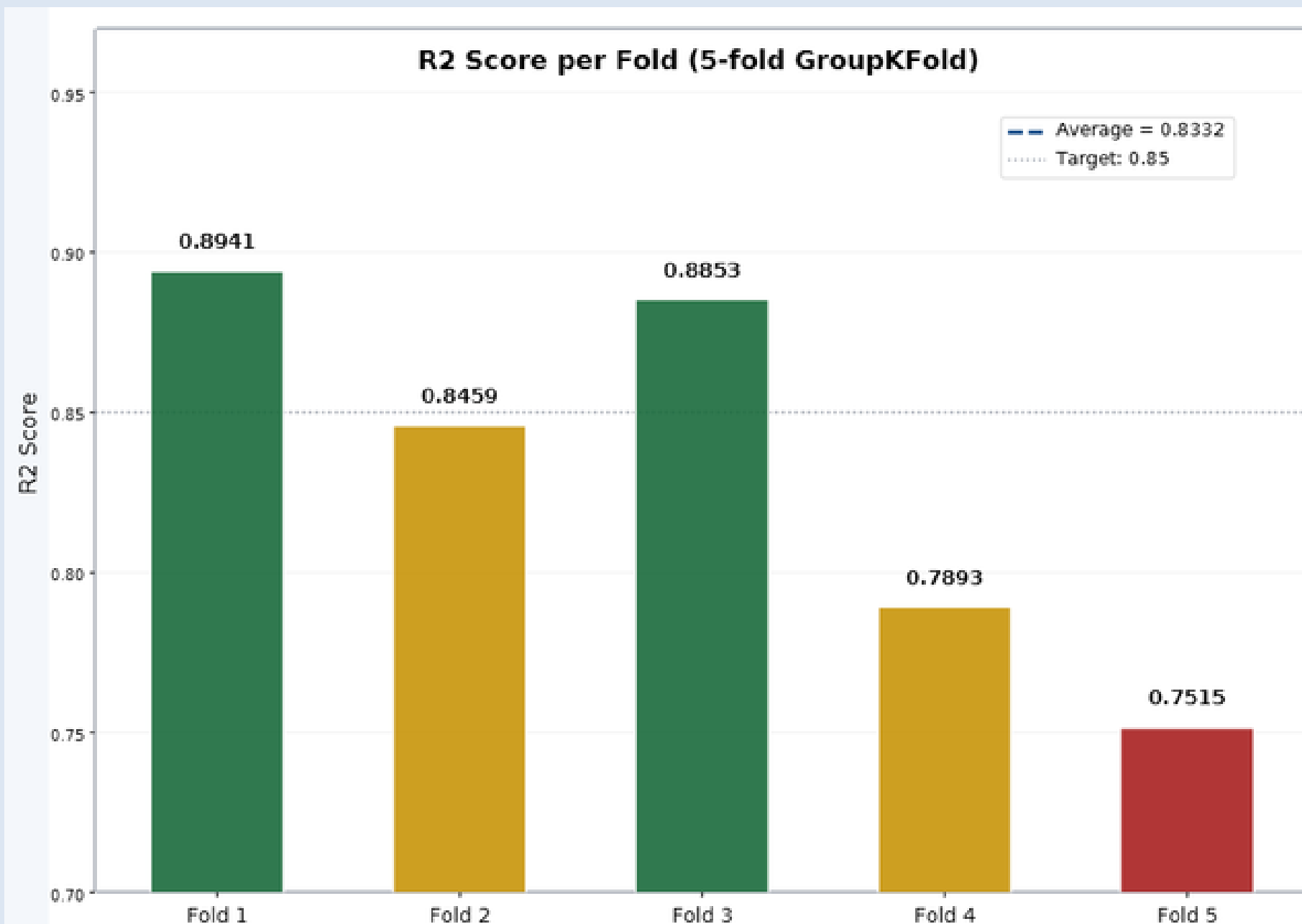
Algorithm Comparison (GroupKFold 5-fold)



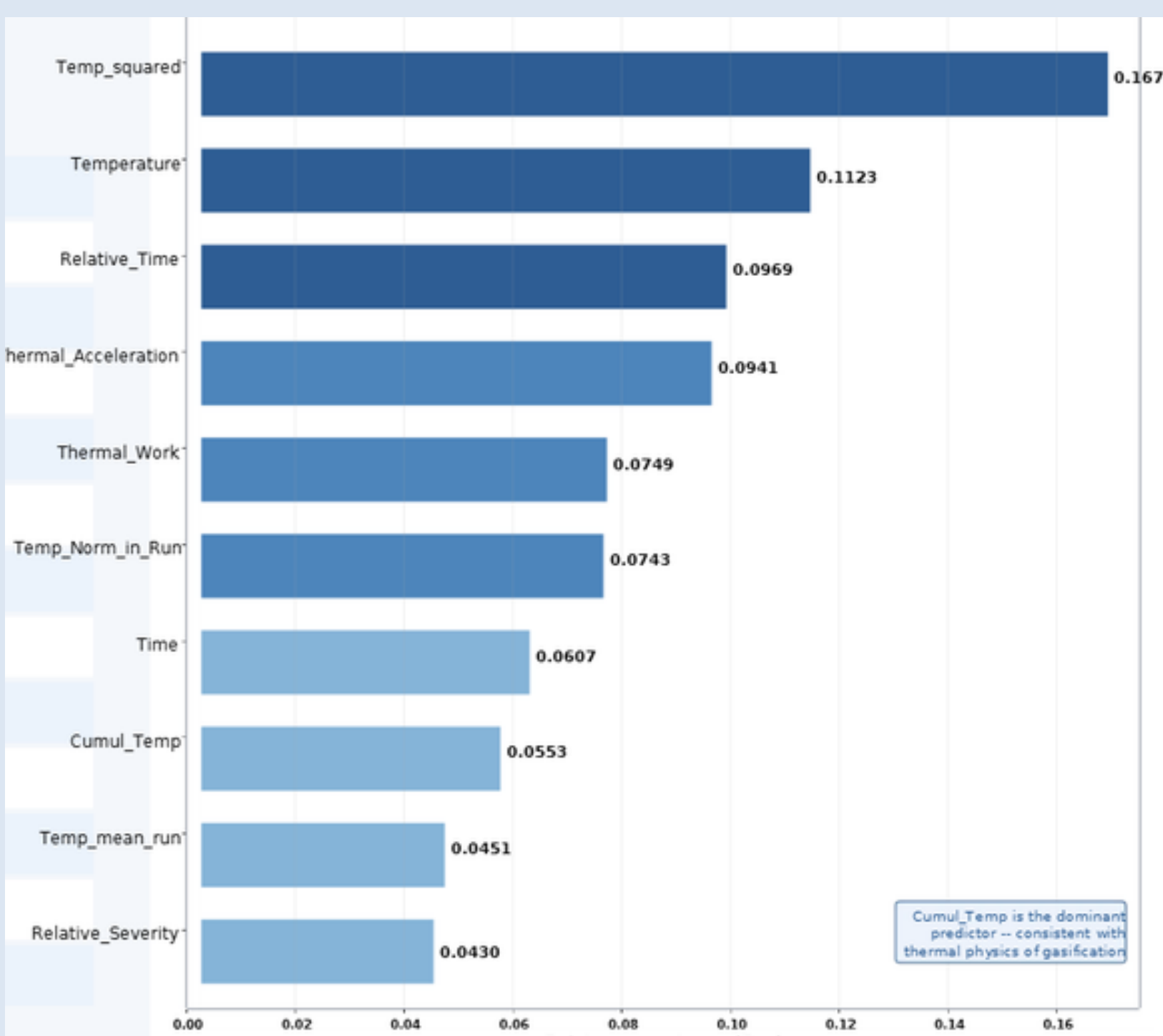
5. CROSS-VALIDATION

GroupKFold keeps each full experiment in one split, avoiding data leakage. 5 folds (global model), 3 folds (Leather specialist).

$R^2=0.828$ test score	$MAE=3.09$ vol%
$RMSE=4.56$ vol%	$StdR^2=0.055$ stable



6. TOP 10 FEATURES



CONCLUSION

ML predicts H_2 gasification yield effectively. The hybrid Extra Trees model offers a real-time, practical alternative to complex physico-chemical models via a Streamlit web app.

7. HYBRID MODEL STRATEGY

Why Two Models ?

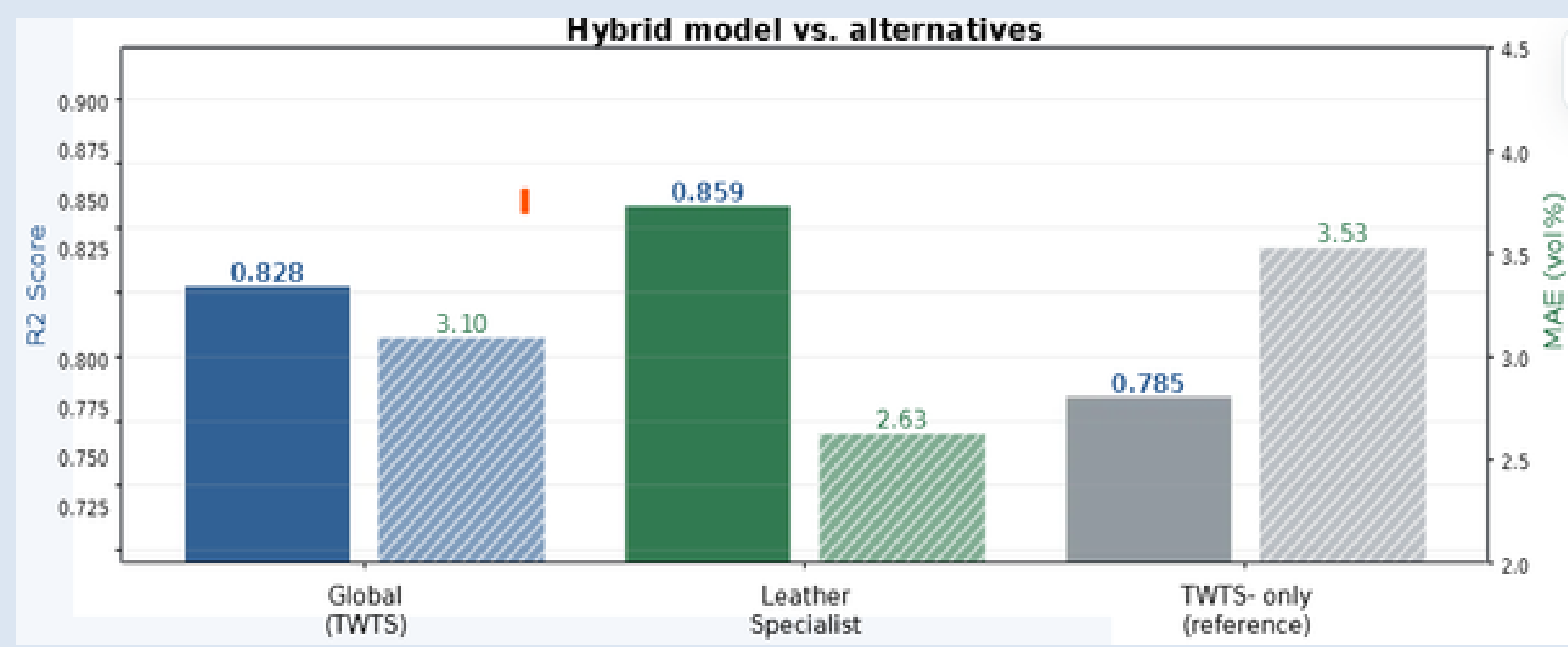
TWTS and Leather scraps have different chemical compositions (cellulose, lignine, proteins, synthetic polymers). A single model makes compromises. We

trained two specialists:

TWTS
model_extra_trees.pkl
 $R^2=0.828$ | $MAE=3.10$

Leather
model_leather.pkl
 $R^2=0.859$ | $MAE=2.63$

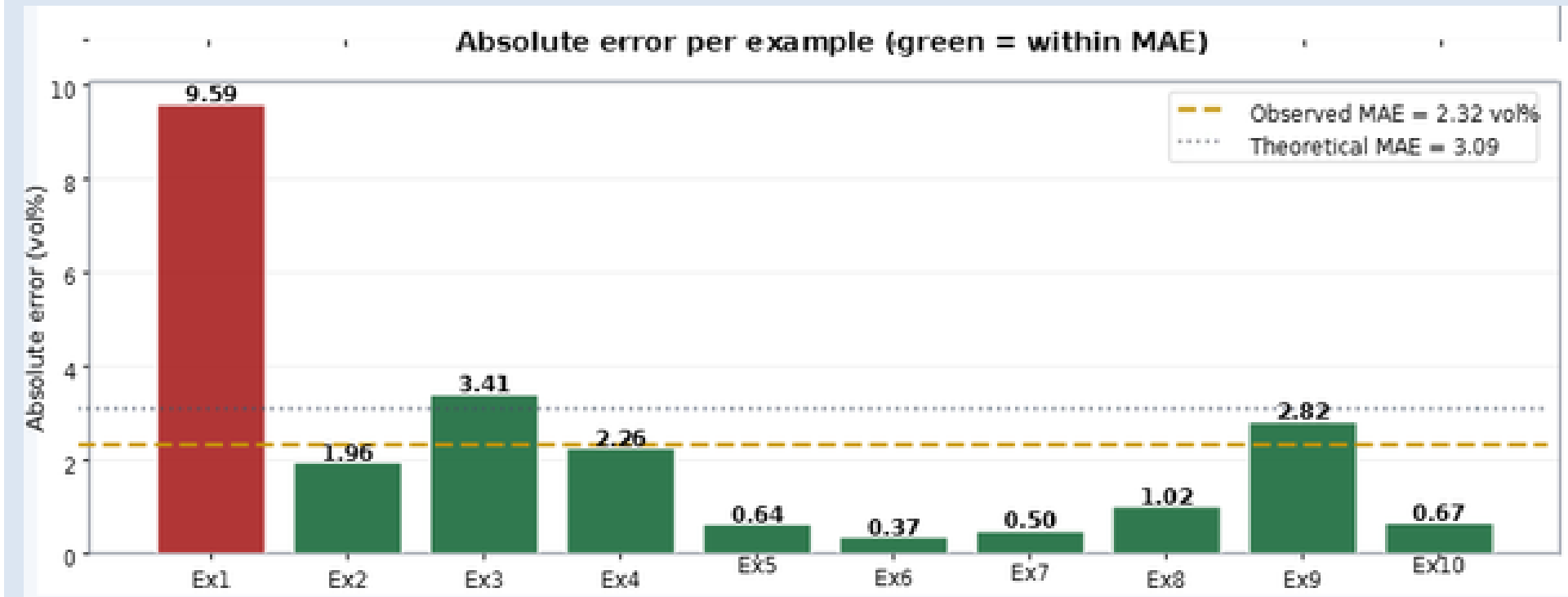
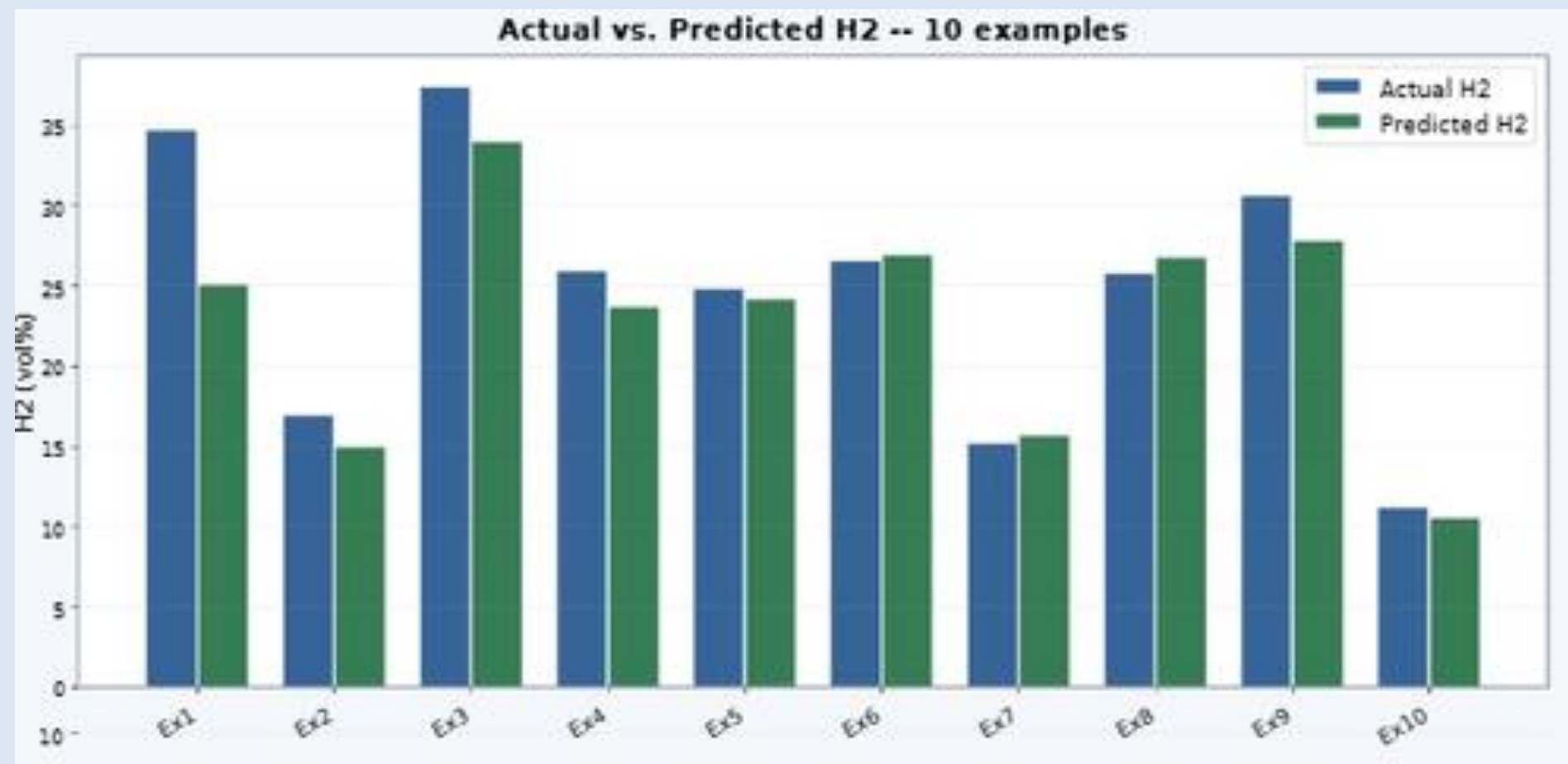
Note: TWTS-only ($R^2=0.785$) is worse than the global model — transfer effect from Leather data.



8. REAL-WORLD VALIDATION

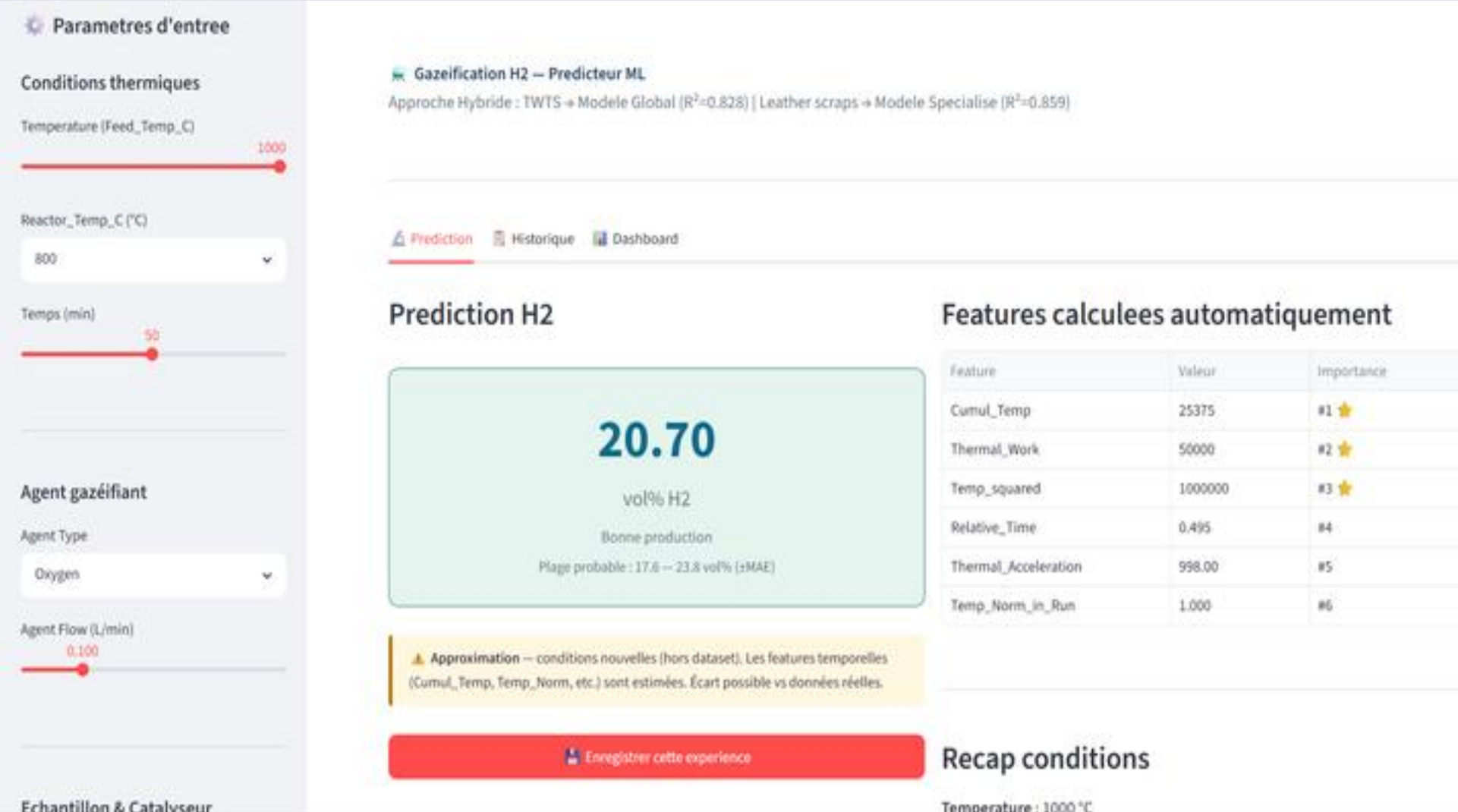
10 examples (5 TWTS + 5 Leather). 9/10 fell within expected error. Obs. $MAE=1.74$ vol% vs theoretical 3.09 vol% (-57%).

9/10 Accurate	1.74 Obs. MAE
0.37 Min error	9.59 Max error



9. WEB APPLICATION

Streamlit app: user inputs conditions $\rightarrow$ instant H_2 prediction.



Future Work

- Expand to 150+ experiments
- Test steam agent to reach $H_2 > 40$ vol%
- Extend to paper, plastic, biomass waste